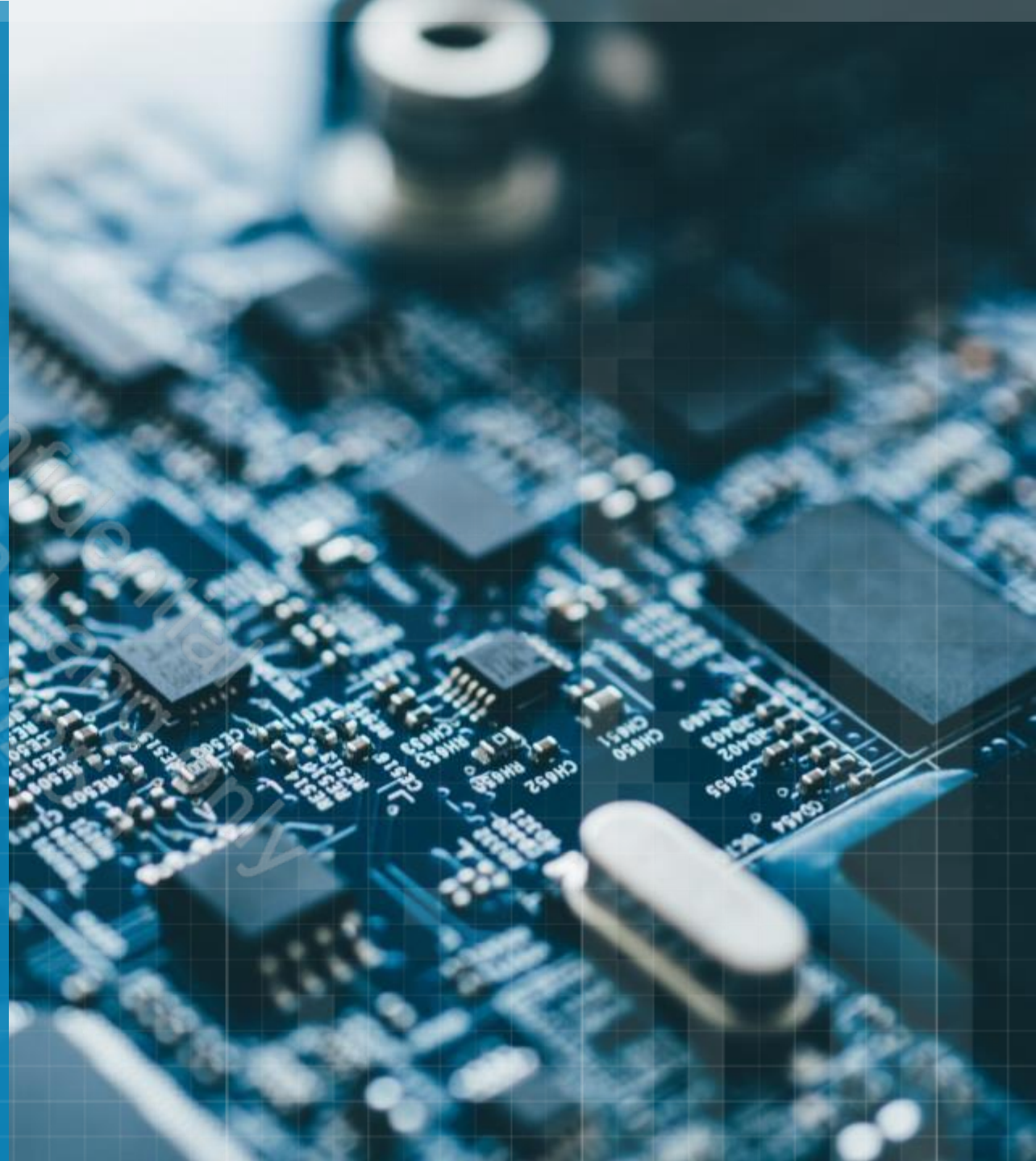
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Image Revert Discussion

Ambarella Taiwan
Duncan Wei, Jasper Wang



Objective

- Topic: How Ambarella can prove that our CE won't produce the revert issue on image?

- Based on Ambarella experiences from other customer projects:

A. Global revert:
- Global related luma values of output YUV are consistent with those of RAW
- Solution:
  Global tone/CE tuning for prevent from revert issue.
- Verification standard:
  Use standard chart to check is there any revert issue.

B. Local revert:
- Local related luma values of output YUV are consistent with those of RAW
- Solution:
  CE/Sharpen tuning for prevent from revert issue.
- Verification standard:
  Use standard chart to check is there any revert issue.

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Input luma image



Global reverted image
(Total level revert)



Input luma image



Local reverted image

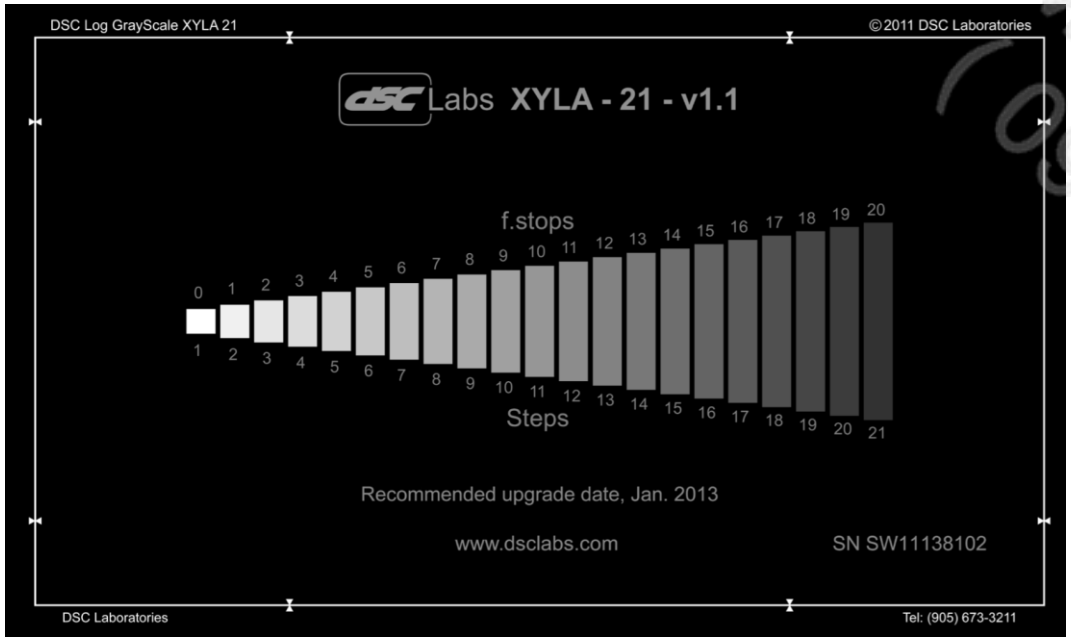


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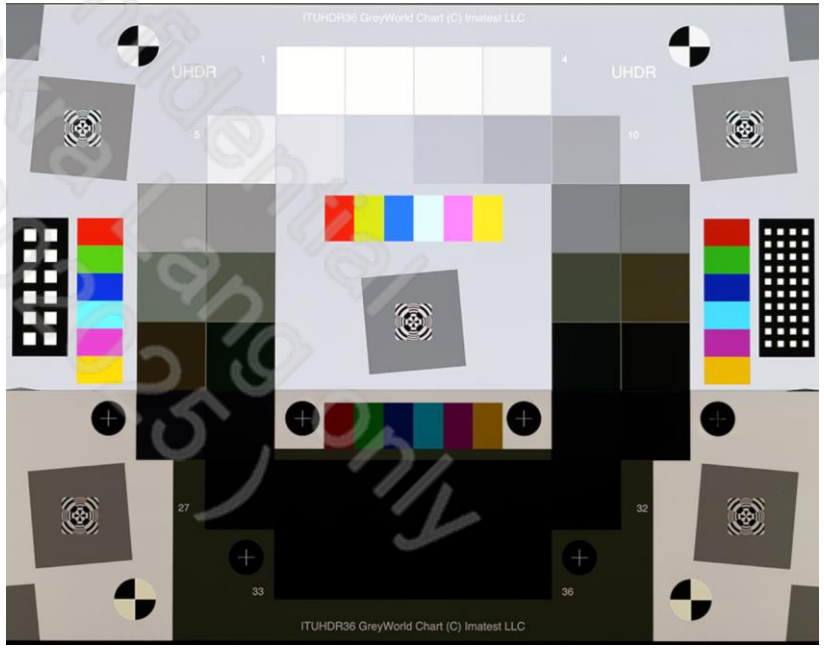
Verification Standard – Test Chart

Testing chart: 21-steps or 36 steps

Imatest Xyla 21 Chart
(Backlight embedded)



Imatest 36-patch Chart
(working with lighting box)





https://www.imatest.com/product/xyla/

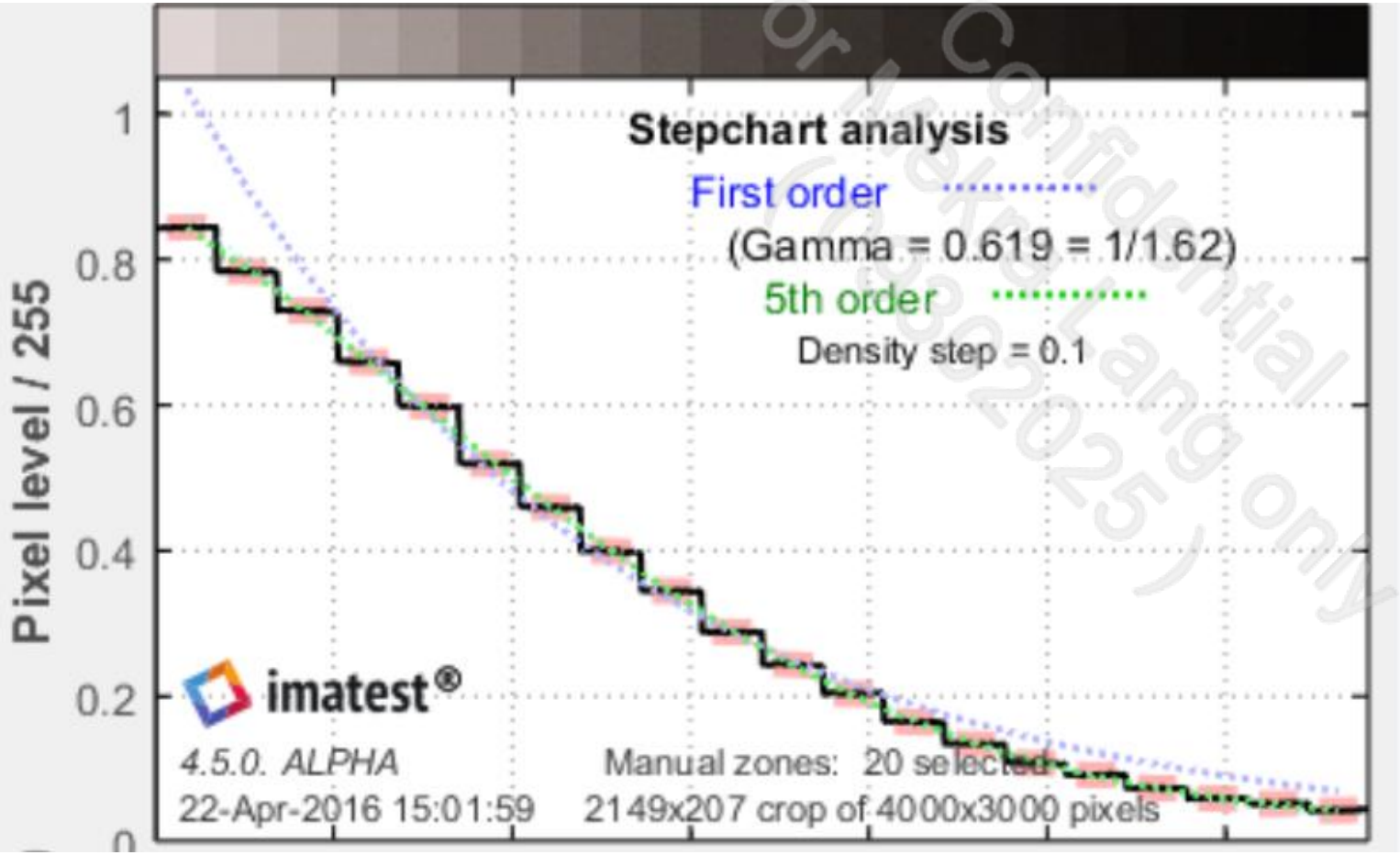
https://www.imatest.com/product/dynamic-range-test-charts/

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Verification Standard – Measurement by Imatest

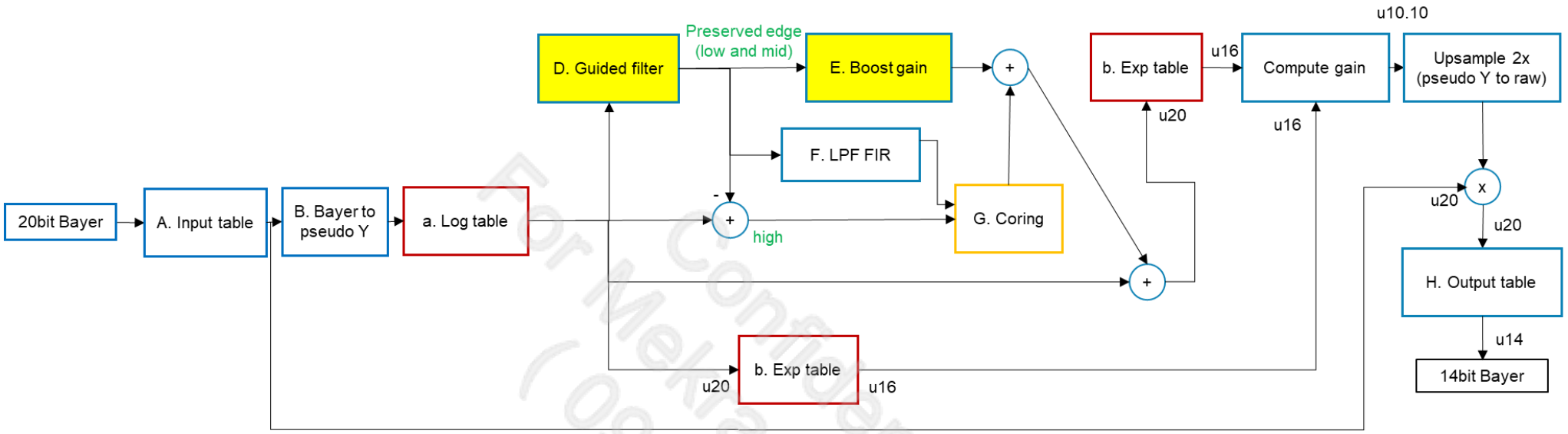
- Measure average value of each patch
- Check there is no any revert issue between each patch
- How can Ambarella ISP minimize revert issue?



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Contrast Enhancement Tuning



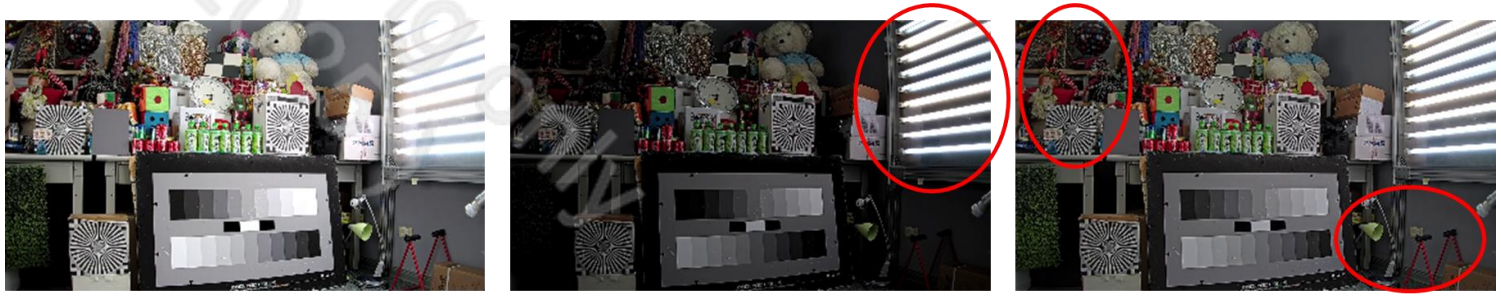
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Guided filter

- Adjustable edge preserving filter, usually for low and mid frequency edge preserving
- Radius and epsilon are two parameters for adjusting edge preserving

Boost gain

- 64 entries boost_table for local tone mapping
- Usually used for boosting shadow detail



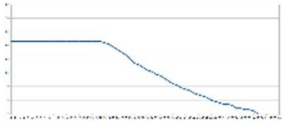
Long/Short Ratio: 2
CE Boost Curve:



Long/Short Ratio: 8
CE Boost Curve:



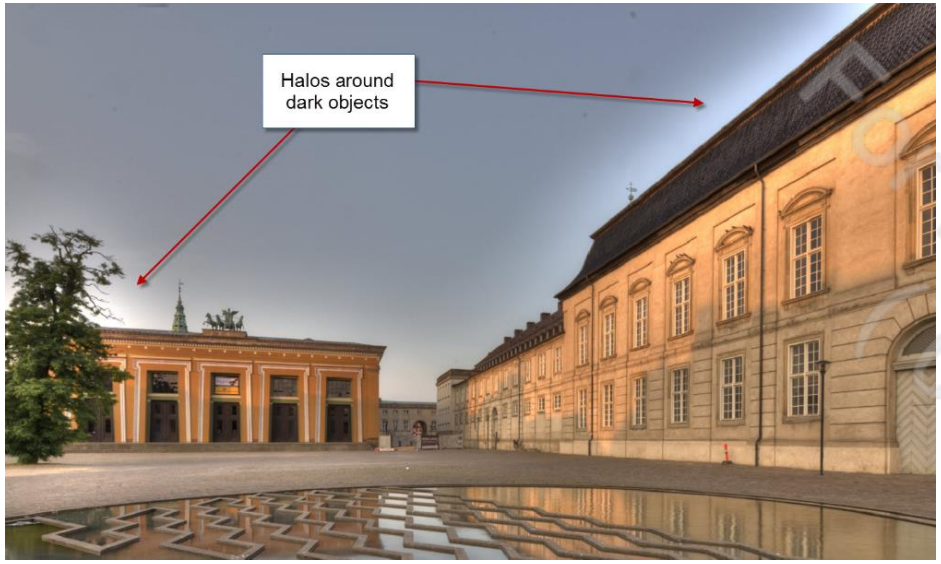
Long/Short Ratio: 8
CE Boost Curve:



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Contrast Enhancement Tuning

- HDR image might add some halo (luma revert) to enhance image contrast





Input & guide

Guided filter
(let $I = p$)
r: local window size



$r = 4, \epsilon = 0.1^2$ $r = 4, \epsilon = 0.2^2$ $r = 4, \epsilon = 0.4^2$

Less halo
Weaker image contrast

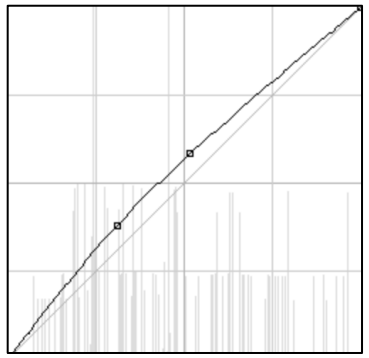
More halo
Stronger image contrast



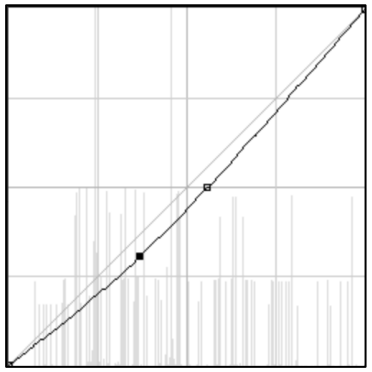
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Global Tone Tuning

- We can apply global tone to match human vison preference.
- Inverse tone curve setting will cause global revert issue.



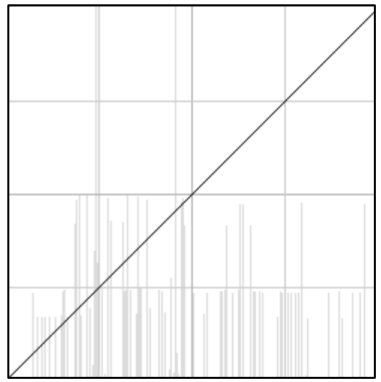
Increase tone



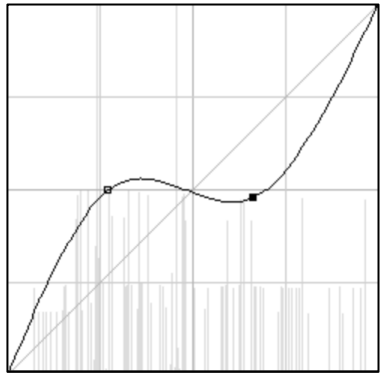
Decrease tone



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Linear tone

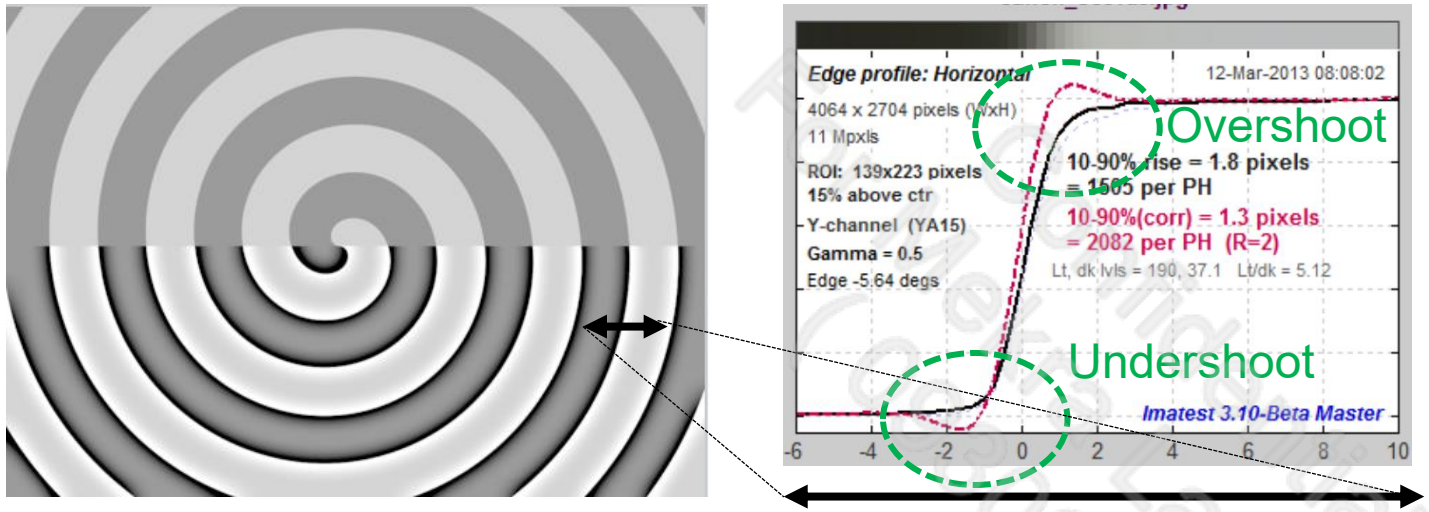


Revert tone

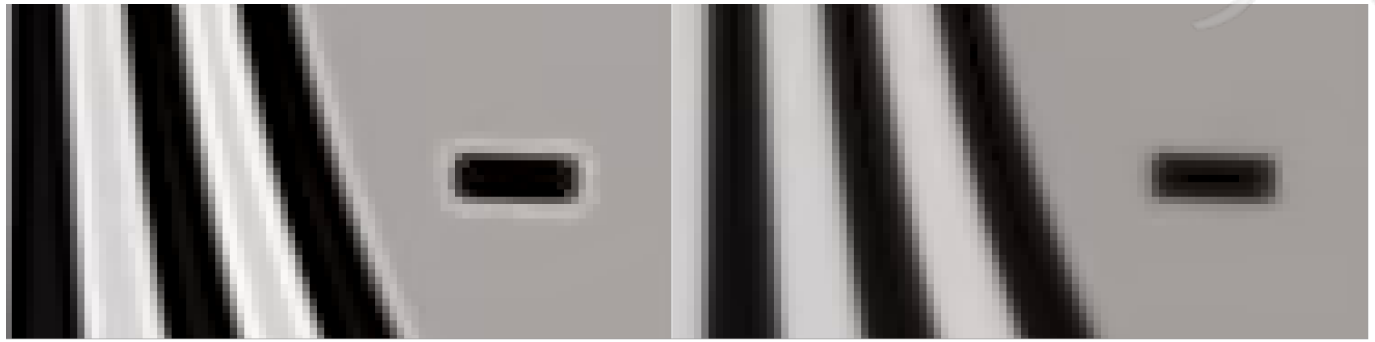
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Sharpness Tuning

- Sharpness may apply overshoot/undershoot to make human vison better, but it will cause image local revert.



- Ambarella sharpen can support overshoot/undershoot detection and eliminate it.



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Conclusions and Discussions

- Ambarella proposed a standard measurement to check revert issue.

- Ambarella ISP can be adjusted to prevent from global/local revert issue.
  - Global tone
  - CE guiding filter/boost
  - Sharpen overshoot/undershoot detection/reduction algorithm

- Could MEKRALANG share some idea of Safety function (revert concert) vs. Image quality?
- For EMR application, some ISP enhancement, such as CE contrast and Sharpen overshoot/undershoot, might be helpful for human viewing.
- However, these image processing may cause “revert” issue, how can we get optimal trade-off?
- Why revert image impact safety function? Maybe some example will be very helpful.

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Example1

Normal image (without revert issue)



High contrast/sharpens image (might with revert issue)



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Example2

Normal image (without revert issue)



High contrast/sharpens image (might with revert issue)



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